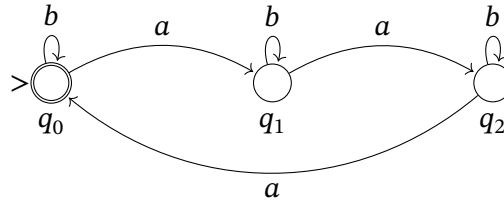


Regular and Nonregular Languages

Theorem (Pumping theorem for regular languages)

Let L be a regular language. There is an integer $n \geq 1$ such that any string $w \in L$ with $|w| \geq n$ can be rewritten as $w = xyz$ such that $y \neq e$, $|xy| \leq n$, and $xy^iz \in L$ for each $i \geq 0$.

Q1. Let M be the DFA below over $\{a, b\}$, accepting the language of all strings in which the number of a 's is divisible by 3.



- (a) Write out the full sequence of configurations attained during $(q_0, baaab) \vdash_M^* (q_0, e)$.
- (b) The proof of the pumping theorem captures the intuition behind it; let us argue in a similar way here. Since $|w| = 5$ and M has $n = 3$ states, the computation of w must visit some state more than once among its first $n + 1$ configurations. Use your answer to (a) to find where this happens, and use it to write $w = xyz$ in the way the pumping theorem needs.
- (c) Verify that xy^0z and xy^2z are both in $L(M)$.

Q2. Explain why you are not free to choose the decomposition xyz yourself when trying to prove some language L is not regular. Illustrate this: pick any nonregular language, and give a string in the language together with a decomposition xyz of it for which xy^iz remains in the language.

Q3. Let $\Sigma = \{a, b\}$, and let $L = \{w \in \Sigma^* : w = xay \text{ for some } x, y \in \Sigma^* \text{ where } |x| = |y|\}$. Using the pumping theorem, show that L is not regular.

Q4. Let $\Sigma = \{a, b\}$, and let

$$L = \{w \in \Sigma^* : \#_b(w) = 2 \cdot \#_a(w)\},$$

where $\#_a(w)$ and $\#_b(w)$ denote the number of a 's and b 's in w .

- (a) Show that $L \cap L(a^*b^*) = \{a^n b^{2n} : n \geq 0\}$.
- (b) Use the pumping theorem to prove that $\{a^n b^{2n} : n \geq 0\}$ is not regular.
- (c) Using closure under intersection and your result from (b), conclude that L is not regular.
- (d) Why is proving that $L \cap L(a^*b^*)$ is not regular enough to conclude L itself is not regular?

Q5. Prove that the following language L over the alphabet $\Sigma = \{a, b, c\}$ is not regular:

$$L = \{ab^n c^n : n \geq 0\} \cup \{a^n b^m c^l : n \neq 1\}$$

Definition: $\approx_L$

Let $L \subseteq \Sigma^*$. Two strings $x, y \in \Sigma^*$ are said to be *equivalent* with respect to L , written $x \approx_L y$, if for every $z \in \Sigma^*$, $xz \in L \iff yz \in L$.

Theorem (Myhill–Nerode Theorem)

A language L is regular if and only if $\approx_L$ has finitely many equivalence classes.

Q6. Let $\Sigma = \{a, b\}$, and let $L = \{w \in \Sigma^* : \#_a(w) \mid 2 \text{ and } \#_b(w) \mid 3\}$.

- (a) Determine whether $b \approx_L bbbb$. Prove your answer.
- (b) Determine whether $a \approx_L ab$. Prove your answer.
- (c) Prove that the language is regular.

Q7. Let $\Sigma = \{a, b\}$, and let $L = \{w \in \Sigma^* : \#_b(w) \geq \#_a(w)\}$.

- (a) Determine whether $abaab \approx_L aab$. Prove your answer.
- (b) Determine whether $a \approx_L aaa$. Prove your answer.
- (c) Is the language regular?
- (d) Prove that L is not regular using the pumping theorem.

Q8. Let $\Sigma = \{a\}$, and let $L = \{a^{n^2} : n \geq 0\}$.

- (a) Prove that L is not regular using the pumping theorem.
- (b) Prove that L is not regular using the Myhill–Nerode Theorem.

Q9. Let $\Sigma = \{a, b\}$, and let $L = \{a^n b^n : n \geq 0\} \cup \{a^m b^n : m \neq n \text{ and } m + n \text{ is odd}\}$.

- (a) Prove that L is not regular using the Myhill–Nerode Theorem.
- (b) Prove that L is not regular using the pumping theorem.